React

The library for web and native user interfaces

Learn React API Reference

Introduction

Agenda

- Background.
- The V in MVC
- TSX (Typescript Extension Syntax).
- Developer tools..
- React Component basics.
- Material Design.

React

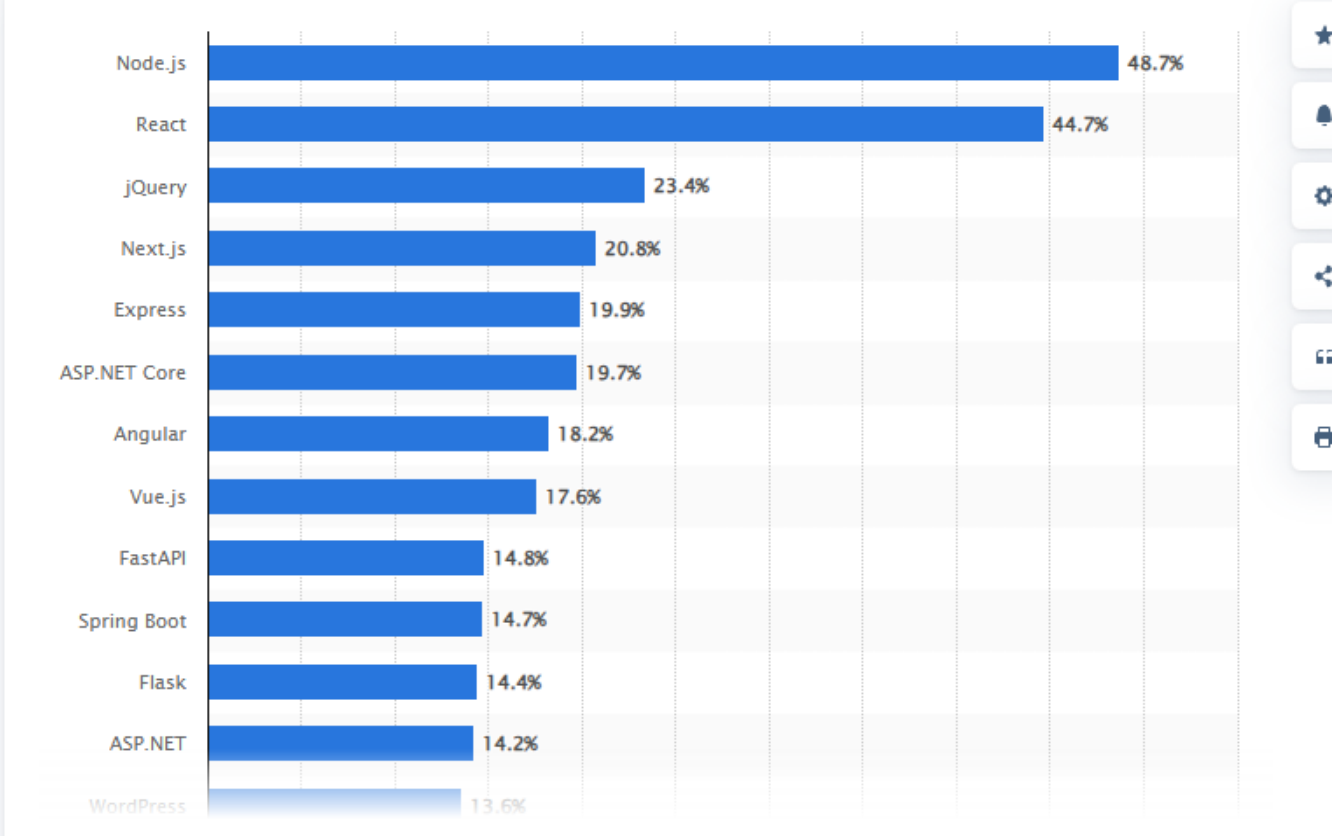
- **A Javascript framework for building dynamic Web** User Interfaces**.**
  - **A Single Page Apps technology.**
  - **Open-sourced in 2012.**

- **Client-side framework.**
  - **More a library than a framework.**

Why React?

Technology & Telecommunications › Software

Most used web frameworks among developers worldwide,



The most widely used web framework/library in the world today is generally considered to be React.

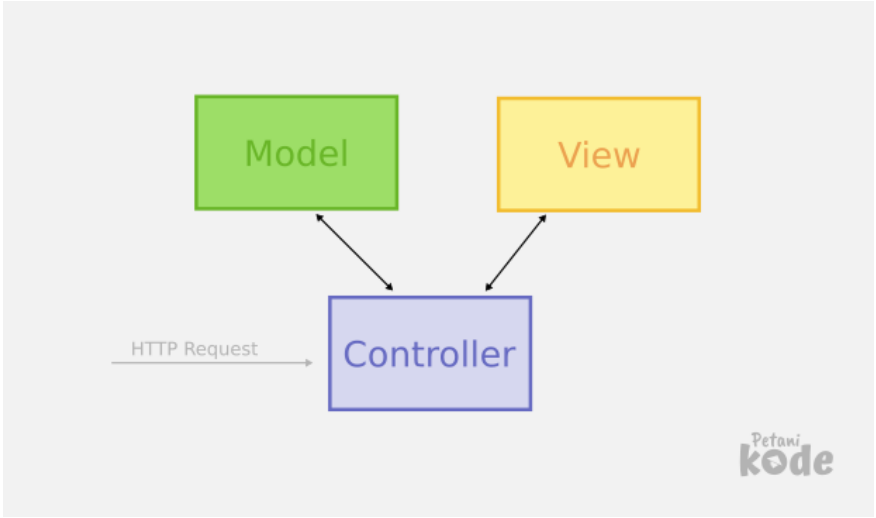
It dominates frontend web development and is used by companies like Meta, Netflix, Airbnb, and many others. Surveys such as Stack Overflow's developer survey consistently rank React at or near the top for web technologies. Strapi +1

Other very popular web frameworks include:

- Angular — common in large enterprise apps Wikipedia
- Vue.js — popular for simplicity and fast learning
- Next.js — growing very rapidly
- Django — widely used for backend/web apps
- Laravel — dominant in PHP ecosystems
- Express.js — very common backend framework

Before React

- MVC pattern **– The convention for** app design. **Promoted by market leaders**, e.g. **AngularJS (1.x), EmberJS, BackboneJS.**
- **React is not MVC, just V.**
  - **It challenged established best practice (MVC).**
- Templating widespread use in the V layer.
  - **React based on “components”.**

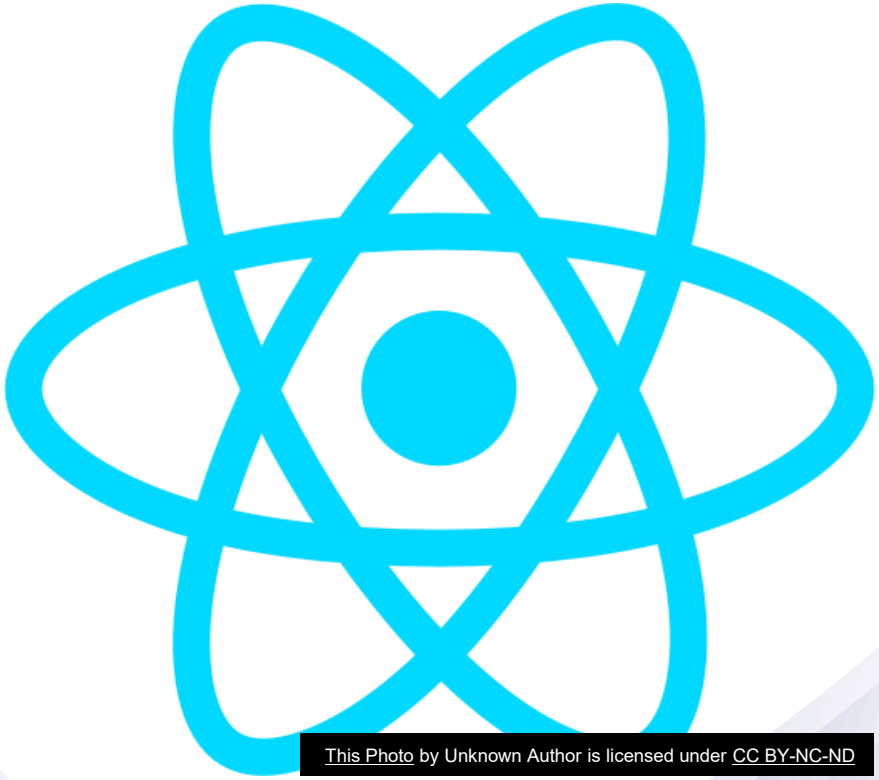


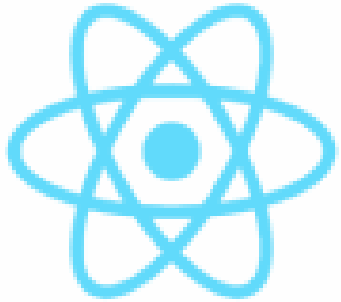
React Components

- **Philosophy:** *Build components, not templates.*
- **All about the User Interface (UI).**
  - Not focused on business logic or the data model (MVC)

- **Component - A unit comprised of:**

  UI description (HTML) + UI behaviour (JS)
  - **Two aspects are tightly coupled and co-located.**
    - Pre-React frameworks decoupled them.
  - **Benefits:**
    1. Improved Composition.
    2. Greater Reusability.

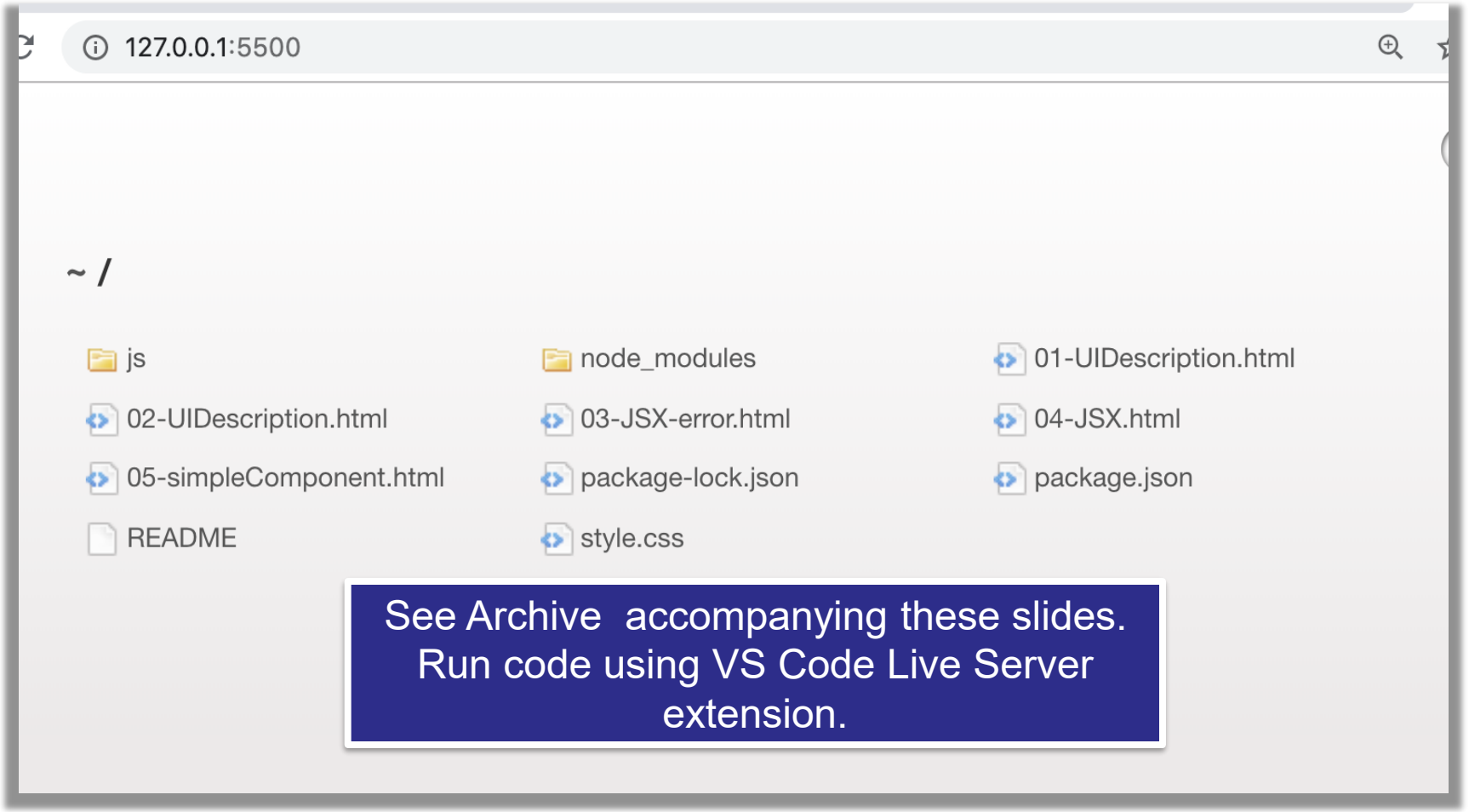


TypeScript  React

- Used to add type definitions to JavaScript codebases.
- TypeScript supports JSX (=>TSX)
- Include in your React project using **@types/react** and **@types/react-dom**
- Needs to be **Transpiled/Compiled** to Javascript to run in Browser/Client.

In-Class Code Demos.

(See lecture archive)



Creating the UI Description.

(Vanilla React)

- React.createElement() **– creates a HTML element.**
- ReactDOM.createRoot() – **used to attach the created element to existing DOM node.**
- React.createElement() takes three or more **arguments**:

Type (e.g. h1, div, …)

Properties (style, event handler…)

Children(0 to many child elements)

```
<div id="mount-point"></div>
<script src="../node_modules/react/umd/react.development.js"></script>
<script src="../node_modules/react-dom/umd/react-dom.development.js"></script>
<script>
  // Create elements imperatively with React.createElement
  const header = React.createElement("h1", { className: "heading" }, "Languages");
```

  - **We don't use createElement() directly - too cumbersome. More later…**
- ReactDOM.createRoot () has 1 **argument:**

DOM element on which to render your React Root Element

```
// Render the elements
const rootElement = ReactDOM.createRoot(document.getElementById("mount-point"))
rootElement.render(root);
```

UI Description Implementation

(the imperative way)

- **See the demos:**
  - **Ref.** 01-UIDescription.html.

  - **Nesting** createElement**() calls (Ref**. 02-UIDescription.html)

- **Imperative programming** is a programming paradigm that uses statements that change a program's state.

  focuses on describing how a program operates, step by step.

- **Declarative programming** is a programming paradigm … that expresses the logic of a computation without describing its control flow.

  focuses on what the result should be without specifying how it should achieve the results

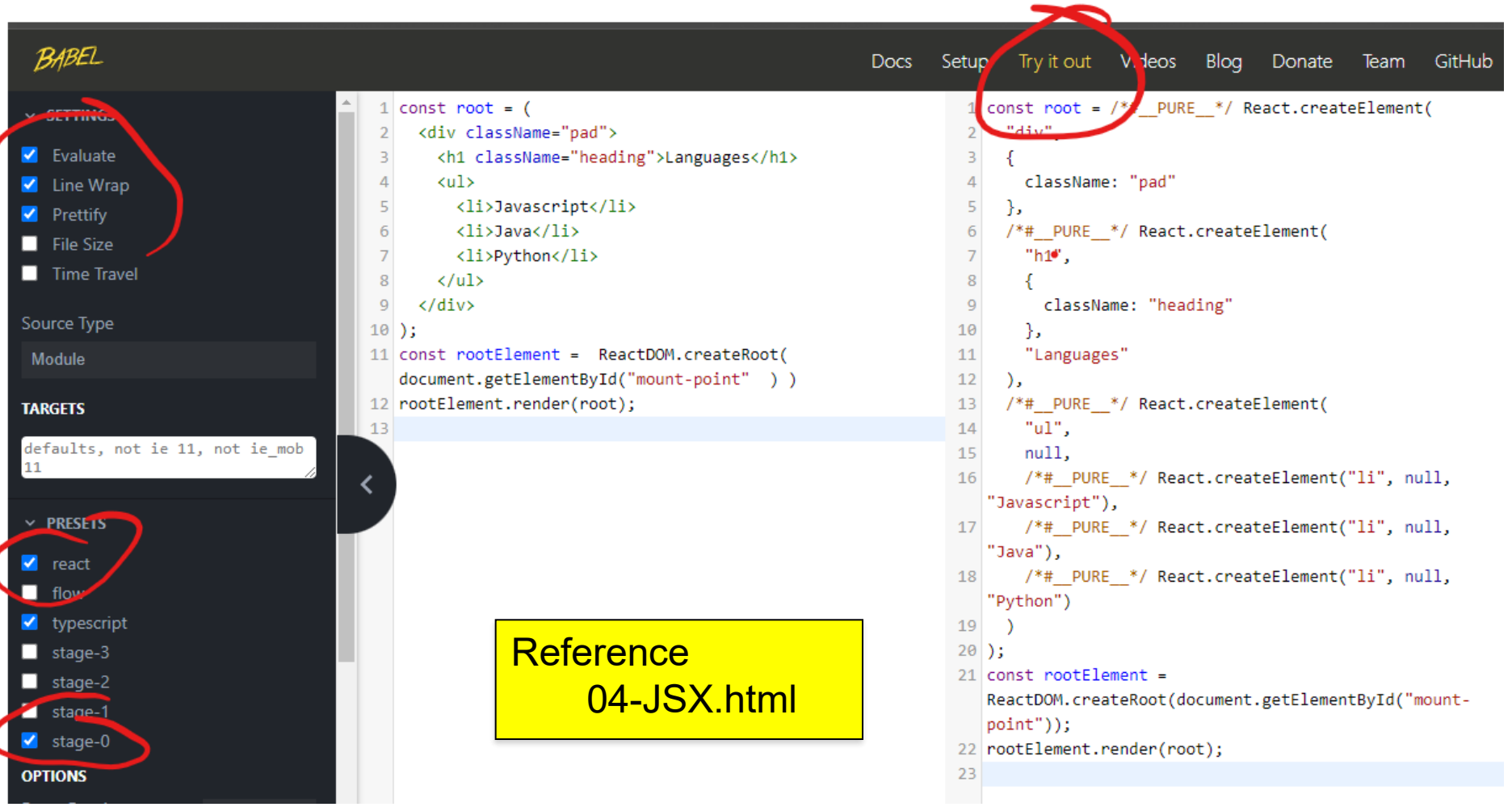
UI description implementation

(the declarative way)

- **TSX – TypeScript XML.**
- Declarative syntax for coding UI descriptions.
- Retains the full power of Typescript.
- Allows tight coupling between UI behavior and UI description.
- **However, must be transpiled before being sent to browser.**
- **Reference** 03-JSX-error.html and 04-JSX.html

```
const root = (
  <div className="pad">
    <h1 className="heading">Languages</h1>
    <ul>
      <li>Javascript</li>
      <li>Java</li>
      <li>Python</li>
    </ul>
  </div>
);
const rootElement =  ReactDOM.createRoot( document.getElementById("mount-point"
rootElement.render(root);
```

REPL (Read-Evaluate-Print-Loop) transpiler.



Reference
04-JSX.html

TSX(TypeScript XML)

- **JSX(Javascript XML)** is a file syntax extension used by React. JSX resembles HTML, allowing developers to write UI components with an HTML-like (it is actually XML).
- **TSX is the TypeScript version of JSX,** TypeScript is a superset that adds static typing in JavaScript.
  - Typescript files containing TSX use the .tsx extension.
- Some minor HTML tag attributes differences, e.g. className (class), htmlFor (for).

```tsx
const App: React.FC<AppProps> = ({ title }) => (
  <div className="app-container">
    <h1 className="app-header">{title}</h1>
    <button onClick={() => console.log('Button clicked!')}>
      Click me
    </button>
  </div>
);
```

- Allows UI description to be coded **in a declarative style** and be in-lined in TypeScript.

- **Combines templating with the power of TS.**

TSX Transpiling

- **So browsers don't do Typescript!**
  - Needs to be **Transpiled to Javascript**
- What can we use to Transpile?
  - The Babel platform.
  - The Vite library.

- How?
  1. Manually, via REPL environment or command line.
     - When experimenting only.
  2. Using an instrumented web server – Vite library instrumentation.
     - Ideal for development.
  3. Using bundler tools as part of the build process – Vite again.
     - Production standard.

React Components.

- **We develop COMPONENTS.**
  - A TS function that returns a UI description, i.e. TSX**.**

- **We reference a component like a HTML tag.**
  e.g.

```
const rootElement =
    ReactDOM.createRoot(document.getElementById("mount-point"));
rootElement.render( <DynamicLanguages/> );
```

- **Reference** 05-simpleComponent.html